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22 May 1958
SUPERSEDING
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MILITARY SPECIFICATION

FIRE CONTROL SIGHTING EQUIPMENT IN HOWITZER,
SELF-PROPELLED, FULL TRACKED, 105MM, M52;
REQUIREMENTS FOR

1. SCOPE

1.1 Scope. - This specification covers the primary and secondary fire control sighting equipment assembled in the Howitzer, Self-propelled, Full Tracked, 105MM, M52.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS
Military

MIL-H-10640	- Howitzer, 105MM, Self-propelled, T98E1 (W/E) (M52).
MIL-I-11748	- Interference Reduction for Electrical and Electronic Equipment.

ENGINEERING PARTS LIST
Ordnance Corps

7333216	- Howitzer, 105MM, Self-propelled, M52.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design. - The fire control sighting equipment shall conform to the drawings specified on Engineering Parts List (EPL) 7333216.

3.1.1 Radio interference suppression. - When specified (see 6.4) equipment procured under this specification shall comply with the class II requirements of Specification MIL-I-11748 when motors capable of producing continuous radio interference are used. Induction or repulsion-induction type motors without commutation devices or those utilizing brush-lifting devices making electrical contact only during the starting cycle are exempt from radio interference treatment or test requirements.

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3.2 Environmental requirements. - When tested as specified in 4.4 through 4.5, the fire control sighting equipment shall be operable in an ambient air temperature of minus 40°F. to plus 125°F.

3.2.1 Storage. - The fire control sighting equipment shall be operable at any point within the temperature range specified in 3.2 after storage for 24 hours in an ambient air temperature of minus 80°F. and plus 160°F.

3.3 Performance requirements.

3.3.1 Weapon stops. - When checked as specified in 4.3, the weapon shall engage stops at elevations of minus 178 to minus 160 mils and plus 1157 to plus 1175 mils.

3.3.2 Mount telescope T179 and T179E4 deviation. - When tested as specified in 4.4.1, the maximum allowable deviation of the vertical cross-hair of the telescope reticle from the target point is 0.7 mil.

3.3.3 Levelness of the azimuth table. - When tested as specified in 4.4.2, the level bubbles of the cross leveling adjustments shall remain centered within plus or minus 1/2 vial graduation. One graduation equals 0.1 inch.

3.3.4 Horizontal deflection of panoramic telescope. - When tested as specified in 4.4.3, the maximum allowable difference in readings after pressures have been applied to the eyepiece in opposite directions is 0.4 mil for zero cant and zero elevation and 2.5 mils for zero cant and 1100 mils elevation. These tolerances shall apply at any position of the azimuth table.

3.3.5 Accuracy of elevation counter. - When tested as specified in 4.4.4, the maximum allowable deviation of the quadrant elevation from the elevation counter setting is 0.5 mil at zero counter setting and 1.0 mil at all other counter settings.

3.3.5.1 Backlash. - When tested as specified in 4.4.4.1, backlash shall be not more than 1.0 mil at any elevation counter reading from minus 160 mils to plus 1157 mils.

3.3.6 Boresight check of panoramic telescope mount. - When tested as specified in 4.4.5, the line of sight of the panoramic telescope shall be within ± 3.0 mils of the target aiming point.

3.3.7 Direct sighting telescope mount. - When tested as specified in 4.5.1, the maximum allowable deviation of the line of sight from a plumb line is 1.0 mil.

3.3.8 Boresight check of direct fire telescope. - When tested as specified in 4.5.2, the line of sight of the direct fire telescope shall be within ± 5.0 mils of the target aiming point.

3.3.9 Boresight retention. - When tested as specified in 4.6, the panoramic telescope line of sight shall have maintained its original setting with respect to the weapon tube in azimuth within ± 0.75 mil and its original setting of elevation counter with respect to the quadrant pads of the breech within ± 0.5 mil. The direct sighting telescope mount shall have maintained its original setting with respect to the tube within ± 0.5 mil in azimuth and ± 0.5 mil in elevation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Tests

4.1.1 Environmental tests. - To determine conformance to 3.2, the fire control sighting equipment shall be tested in accordance with the procedures specified in 4.4 and 4.5. Tests shall be performed at a Government proving ground.

4.1.2 Acceptance tests. - To determine conformance to 3.3, the fire control sighting equipment shall be tested in accordance with the procedures specified in 4.4 through 4.6. Tests shall be conducted at existing air temperatures.

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4.2 Equipment. - The following fire control test equipment shall be used:

(a) Mechanical jacks or a leveling platform for leveling the vehicle.
(b) A plumbline, 1/16 inch in diameter shall be suspended 15 feet from the muzzle of the weapon. The line shall be shielded from wind currents and the plumb bob shall be immersed in a container of oil to act as a dampener.
(c) A boresight retention and checking target shall be constructed of non-warping and non-shrinking material designed as shown in figure 1. The target shall be placed at a distance of not less than 50 feet from the center of the weapon trunnions in a plane perpendicular to the weapon bore at zero weapon elevation. The reference line on the target shall be plumb. The height of the target above ground shall be such that at zero weapon elevation the center line of the weapon bore shall fall on the weapon aiming point within ± 1/8 inch.
(d) An approved "borescope" (see 6.3) shall be used. This shall be a telescope with a crosshair reticle mounted in a self alining fixture to be inserted in the breech of the weapon so that the crosshairs of the telescope fall on the center line of the bore. The scope shall have a separate focus for reticle and field and be adjustable from 5 yards to 5000 yards. The reticle of the telescope shall be centered with respect to the outside diameter of the adapter within 0.1 mil. The scope shall have a magnification of 20 X or higher.
(e) Parallax shields with a centrally located 1/16-inch maximum diameter hole shall be placed over the eyepiece of the telescope to eliminate parallax.
(f) A master Panoramic Telescope T149E1, a master Direct Sighting Telescope T150E1 and a master T38 Periscope. A master telescope or periscope is defined as a scope with the line of sight fixed in correct relationship to the mounting surfaces.
(g) A calibrated gunner's Quadrant M1. A calibrator gunner's quadrant is defined as a quadrant that has been checked on a sine plate at each of the elevations for which the quadrant will be employed during the test and the deviation from true elevation recorded in a table which will be furnished with the quadrant. The readings of the quadrant shall be corrected accordingly when checking specification tolerances.

4.3 Preliminary checks and adjustments. - The following preliminary checks and adjustments shall be made:

(a) The mounts shall be inspected for tight mechanical installation and proper pinning.
(b) No interference shall exist between or among components of the fire control and components of the vehicle at or between the maximum limits of elevation.

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(c) The vehicle shall be cross leveled so that the weapon will track the plumb line within the thickness (1/16-inch) of the line. This shall be done by using the borescope (see 4.2(d) inserted in the breech and setting the muzzle cross wires on the center of the plumb line at zero weapon elevation. The weapon shall be elevated to 30 degrees and the above tolerance checked. The vehicle shall also be leveled fore and aft so that at zero weapon elevation the weapon will remain at zero ± 0.1 mil when the turret is traversed 30 degrees to the right or left of center. The above shall be checked with a calibrated gunner's quadrant on the breech pads of the weapon. Cross leveling of the vehicle shall be checked periodically throughout the test.

(d) The weapon stop settings shall be checked by the calibrated gunner's quadrant to ascertain conformance to 3.3.1.

(e) The azimuth feedback sector shall be observed as the cab is traversed to its maximum limits left and right, to ascertain that the sector does not run off the pinion.

(f) The micro-switch circuit shall be checked to ascertain that the switch opens when the bell crank opens. This check shall apply to the T179 Telescope Mount only.

4.4 Test for mounts, T179 and T179E4.

4.4.1 Deviation. - The master panoramic telescope shall be placed in the socket and locked in place. A parallax shield shall be placed over the eyepiece. The cross leveling mechanism shall be adjusted to give least movement of the line of sight of the panoramic telescope from a target point through 360 degrees rotation of the azimuth table. At zero elevation, the vertical cross hair of the panoramic telescope shall be placed on the aiming point. The weapon shall be elevated from zero to 1100 mils and checked for conformance to 3.3.2.

4.4.2 Levelness of azimuth table. - Without altering the cross leveling adjustments made in 4.4.1, the azimuth table shall be rotated through 360 degrees and checked for conformance to 3.3.3.

4.4.3 Horizontal deflection of panoramic telescope. - A gradual steady pull of 50 pounds shall be applied to the eyepiece of the telescope, tangentially to the right and perpendicular to the axis of the telescope, and then gradually released. While sighting through the telescope, the vertical cross hair shall then be placed on the center of the panoramic telescope aiming point. A pull of 50 pounds shall be applied to the eyepiece in the opposite direction and gradually released. The deflection of the vertical cross hair relative to the aiming point shall be checked for conformance to 3.3.4.

4.4.4 Accuracy of elevation counter.- A calibrated gunner's quadrant shall be placed on the breech pads of the weapon and the difference between counter reading and weapon elevation shall be checked and noted. The check points shall include minus 155, zero, 225, 625, 1025, and 1155 mils counter elevation. At each setting, the counter indexes shall be made coincident by laying the weapon from low to high to exclude backlash and true weapon elevation shall be read, by the gunner's quadrant, to ascertain conformance to 3.3.5.

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4.4.4.1 Backlash. - The elevation counter indexes shall be made coincident by laying the weapon from low to high and reading weapon elevation by use of the gunner's quadrant. The elevation counter indexes shall again be made coincident by laying the weapon from high to low and again reading the gunner's quadrant. The difference between the two readings is the backlash at the given elevation and shall be noted for conformance to 3.3.5.1.

4.4.5 Boresight check of panoramic telescope mount. - The weapon shall be placed on its aiming point on the boresighting retention and checking target by viewing through the bore (see 4.2(d). With the master panoramic telescope installed in the mount, position shall be noted to ascertain conformance to 3.3.6.

4.5 Test for mount T169E1.

4.5.1 Alinement of telescope. - A T150E1 Telescope and a T38 Periscope shall be placed in the T169E1 Mount and locked in place. The movement of the mushroom stud in the clearance slot shall not exceed 0.020 inch in any direction as the weapon moves through its full range of elevation and depression. Viewing through the telescope with the parallax shield installed over the eyepiece, the line of sight shall be set on the plumb line at zero elevation of the weapon. The weapon shall be elevated through 30 degrees elevation and position noted to ascertain conformance to 3.3.7.

4.5.2 Boresight check of direct fire telescope. - The weapon shall be placed on its aiming point on the boresight retention and checking target. With the master T150E1 Telescope and master T38 Periscope locked in place, location of the line of sight on the target shall be noted to ascertain conformance to 3.3.8. The electrical system shall be checked for illumination of the reticle and cant corrector level vial.

4.6 Boresight retention. - Prior to the 10-mile road test outlined in Specification MIL-H-10640, the crosshairs of the panoramic telescope and direct sighting telescope shall be placed on the center of their respective aiming points shown in figure 1 after laying the weapon exactly on the center. Following the 10-mile road test the vehicle shall be returned to the place from which the original settings were made. Approaching the weapon aiming point from the same direction as before the test, the weapon shall be boresighted at proper aiming point on the target and observation shall be made through the panoramic telescope and the direct sighting telescope to ascertain conformance to 3.3.9.

5. PREPARATION FOR DELIVERY

There are no applicable requirements.

6. NOTES

6.1 This specification is intended for use in conjunction with Specification MIL-H-10640.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

6.3 Telescope. - Taylor Hobson No. 112/392, or equal, telescope has been found to be satisfactory for the telescope specified in 4.2 (d).

6.4 Radio interference suppression. - When applicable, the contracting officer should arrange for investigation and test of the suppression components and system by the Signal Corps in accordance with Specification MIL-I-11748.

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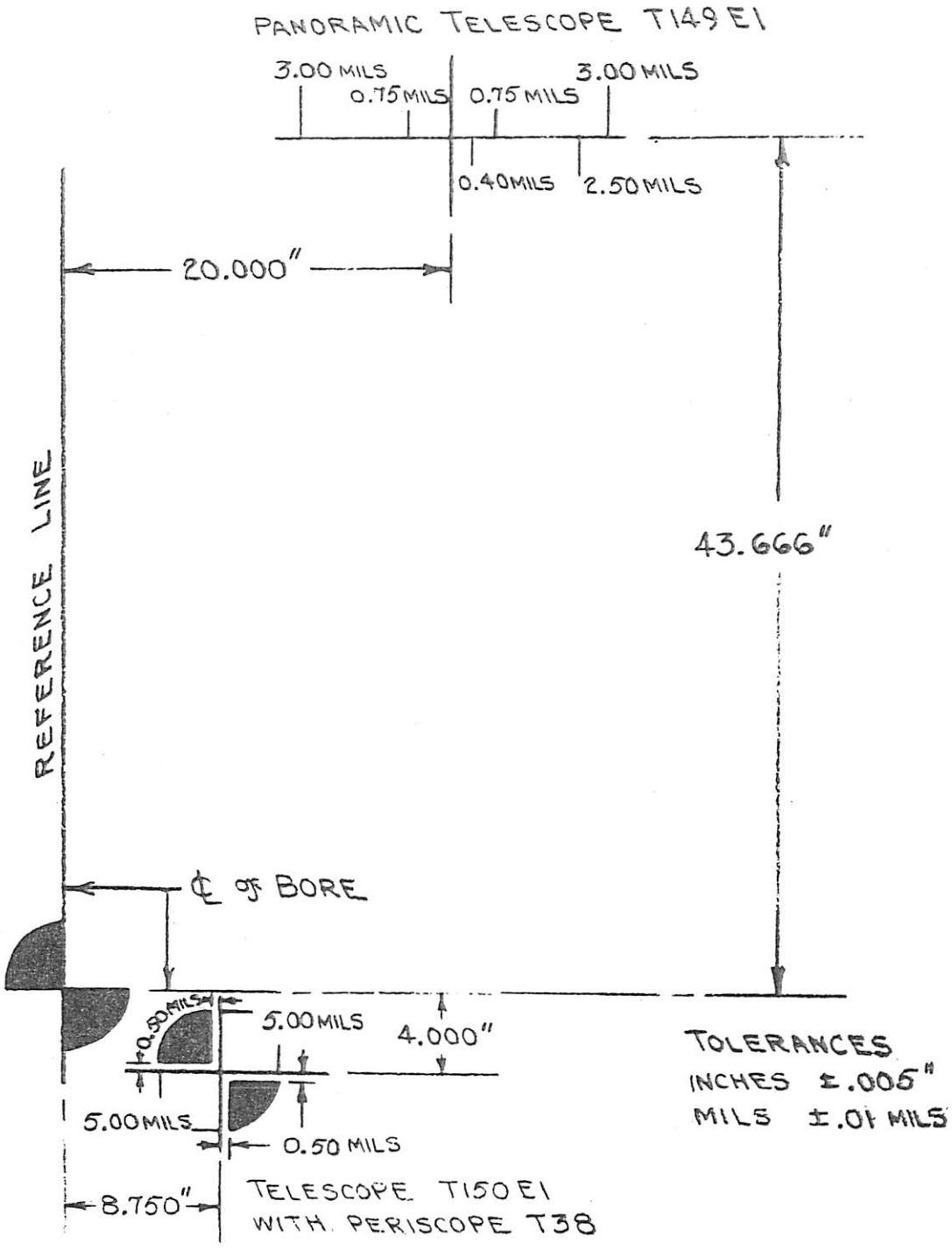


FIG. 1

BORESIGHT RETENTION AND CHECKING TARGET
FOR HOWITZER SELF PROPELLED M52

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